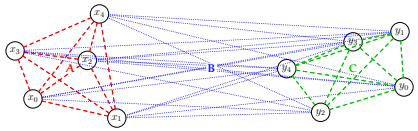
Mathematics of Machine Learning

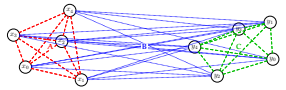
Gaussian Integrals



Gaussian integrals, completing the square, moment generating functions, marginals, conditionals, Kalman filter

Outline

1. One Dimension
2. Many Dimensions
3. Marginals and Conditionals
4. Inference with Gaussians



Why Gaussians?

- Gaussians (normal distributions) are everywhere in machine learning: Gaussian processes, Kalman filters, linear regression, the noise in diffusion models, . . .
- The reason is that we can do all their integrals exactly
- Marginalising, conditioning, adding or linearly transforming Gaussians gives another Gaussian
- The algebra is long-winded, but it uses only three tricks: the basic Gaussian integral, completing the square, and diagonalising a matrix
- Know the results and when to use them; the derivations are there so you can trust them

The Gaussian Integral

- We cannot find an antiderivative of $\mathrm{e}^{-x^2/2}$, but there is a trick: square the integral

$$I^2 = \int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x \int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2}\mathrm{d}y = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)/2}\mathrm{d}x\mathrm{d}y$$

- In polar coordinates ($x^2 + y^2 = r^2$, $\mathrm{d}x\mathrm{d}y = r\mathrm{d}r\mathrm{d}\theta$)

$$I^2 = \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} r\mathrm{e}^{-r^2/2}\mathrm{d}r = 2\pi \int_0^{\infty} \mathrm{e}^{-u}\mathrm{d}u = 2\pi$$

  using $u = r^2/2$, $\mathrm{d}u = r\mathrm{d}r$

- So

$$\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x = \sqrt{2\pi}$$

Normal Distribution

- Changing variables, $z = (x-\mu)/\sigma$ so $\mathrm{d}x = \sigma\,\mathrm{d}z$,

$$\int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}\mathrm{d}x = \sigma\int_{-\infty}^{\infty}\mathrm{e}^{-z^2/2}\mathrm{d}z = \sqrt{2\pi}\sigma$$

- So the normal density

$$\mathcal{N}(x \mid \mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

  integrates to one

- (For small σ the density exceeds one at $x=\mu$: densities are not probabilities)

Completing the Square

- Any quadratic in the exponent can be rewritten (for $a>0$)

$$-\frac{a}{2}x^2 + bx = -\frac{a}{2}\left(x-\frac{b}{a}\right)^2 + \frac{b^2}{2a}$$

- So the integral is a shifted Gaussian times a constant

$$\int_{-\infty}^{\infty}\mathrm{e}^{-\frac{a}{2}x^2+bx}\mathrm{d}x = \mathrm{e}^{\frac{b^2}{2a}}\int_{-\infty}^{\infty}\mathrm{e}^{-\frac{a}{2}\left(x-\frac{b}{a}\right)^2}\mathrm{d}x = \sqrt{\frac{2\pi}{a}}\,\mathrm{e}^{\frac{b^2}{2a}}$$

- This one result does most of the work in this lecture

Moment Generating Function

- The **moment generating function** of X is $M(t) = \mathbb{E}\left[\mathrm{e}^{tX}\right]$

- For $X \sim \mathcal{N}(\mu,\sigma^2)$, completing the square gives

$$M(t) = \int_{-\infty}^{\infty}\mathrm{e}^{tx}\mathcal{N}(x \mid \mu,\sigma^2)\mathrm{d}x = \mathrm{e}^{\mu t+\frac{1}{2}\sigma^2t^2}$$

- Expanding $\mathrm{e}^{tX} = 1 + tX + \frac{1}{2}t^2X^2 + \cdots$ shows $M^{(n)}(0) = \mathbb{E}[X^n]$

$$M'(0) = \mu, \qquad M''(0) = \mu^2 + \sigma^2$$

- The cumulant generating function $\log M(t) = \mu t + \frac{1}{2}\sigma^2t^2$ stops at t^2: a Gaussian has no cumulants beyond the mean and variance

Sums of Gaussians

- A moment generating function determines its distribution

- If X and Y are independent, $\mathbb{E}\left[\mathrm{e}^{t(X+Y)}\right] = \mathbb{E}\left[\mathrm{e}^{tX}\right]\mathbb{E}\left[\mathrm{e}^{tY}\right]$

- So for $X \sim \mathcal{N}(\mu_1,\sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2,\sigma_2^2)$

$$\mathbb{E}\left[\mathrm{e}^{t(X+Y)}\right] = \mathrm{e}^{(\mu_1+\mu_2)t+\frac{1}{2}(\sigma_1^2+\sigma_2^2)t^2}$$

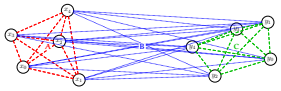
  which is the MGF of $\mathcal{N}(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$

- Means add and variances add, and the sum is still Gaussian

- No integral needed: this is why MGFs are so useful

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2. **Many Dimensions**
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Independent Variables

- With $\boldsymbol{x} = (x_1,\ldots,x_n)^\mathsf{T}$ and $\|\boldsymbol{x}\|^2 = \sum_i x_i^2$, the exponential of a sum is a product of exponentials

$$\int \mathrm{e}^{-\frac{1}{2}\|\boldsymbol{x}\|^2}\mathrm{d}\boldsymbol{x} = \prod_{i=1}^n \int_{-\infty}^{\infty} \mathrm{e}^{-x_i^2/2}\mathrm{d}x_i = (2\pi)^{n/2}$$

- Similarly with a different variance λ_i in each direction

$$\int \mathrm{e}^{-\frac{1}{2}\sum_i x_i^2/\lambda_i}\mathrm{d}\boldsymbol{x} = \prod_{i=1}^n \sqrt{2\pi\lambda_i}$$

- The general case can be rotated into this one

The Multivariate Normal

- For a positive definite covariance $\boldsymbol{\Sigma} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\mathsf{T}$ (eigenvalues $\lambda_i > 0$) consider $I = \int \mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}\mathrm{d}\boldsymbol{x}$
- Rotate, $\boldsymbol{y} = \boldsymbol{V}^\mathsf{T}(\boldsymbol{x}-\boldsymbol{\mu})$: the Jacobian $\boldsymbol{V}$ has $|\det \boldsymbol{V}| = 1$ (rotations keep volumes)

$$I = \int \mathrm{e}^{-\frac{1}{2}\boldsymbol{y}^\mathsf{T}\boldsymbol{\Lambda}^{-1}\boldsymbol{y}}\mathrm{d}\boldsymbol{y} = \prod_i \sqrt{2\pi\lambda_i} = \sqrt{\det(2\pi\boldsymbol{\Sigma})}$$

  as $\det\boldsymbol{\Sigma} = \prod_i \lambda_i$ and $\det(c\boldsymbol{\Sigma}) = c^n \det\boldsymbol{\Sigma}$

- So the multivariate normal density is

$$\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}}\mathrm{e}^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$

Completing the Square in Many Dimensions

- For symmetric positive definite $\boldsymbol{A}$

$$-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}^\mathsf{T}\boldsymbol{x} = -\frac{1}{2}(\boldsymbol{x}-\boldsymbol{A}^{-1}\boldsymbol{b})^\mathsf{T}\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{A}^{-1}\boldsymbol{b}) + \frac{1}{2}\boldsymbol{b}^\mathsf{T}\boldsymbol{A}^{-1}\boldsymbol{b}$$

  (multiply out the right-hand side to check)

- So, by the last slide with $\boldsymbol{\Sigma} = \boldsymbol{A}^{-1}$,

$$\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\boldsymbol{A}\boldsymbol{x}+\boldsymbol{b}^\mathsf{T}\boldsymbol{x}}\mathrm{d}\boldsymbol{x} = \sqrt{\frac{(2\pi)^n}{\det\boldsymbol{A}}}\,\mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\boldsymbol{A}^{-1}\boldsymbol{b}}$$

- This is the central result: every Gaussian integral in this course is an instance of it

Multivariate Moment Generating Function

- For $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the central result with $\boldsymbol{A} = \boldsymbol{\Sigma}^{-1}$ gives

$$M(\boldsymbol{t}) = \mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{\Sigma}\boldsymbol{t}}$$

- Differentiating at $\boldsymbol{t} = \boldsymbol{0}$

$$\mathbb{E}[\boldsymbol{x}] = \boldsymbol{\mu}, \qquad \mathbb{E}\left[(\boldsymbol{x} - \boldsymbol{\mu})(\boldsymbol{x} - \boldsymbol{\mu})^\mathsf{T}\right] = \boldsymbol{\Sigma}$$

- A Gaussian is completely described by its mean and its covariance

Linear Transformations

- Let $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{c}$ with $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- Its MGF is

$$\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{z}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{c}}\mathbb{E}\left[\mathrm{e}^{(\boldsymbol{W}^\mathsf{T}\boldsymbol{t})^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}(\boldsymbol{W}\boldsymbol{\mu}+\boldsymbol{c}) + \frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\mathsf{T}\boldsymbol{t}}$$

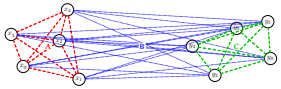
- So

$$\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{W}\boldsymbol{\mu} + \boldsymbol{c},\, \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\mathsf{T})$$

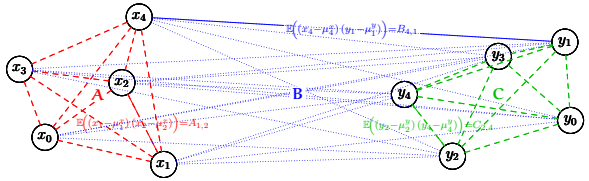
- Linear maps of Gaussians are Gaussian, and we only need to track the mean and covariance ($\boldsymbol{W}$ need not be square)

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Joint Gaussians



$$\begin{pmatrix} \boldsymbol{z}_1 \\ \boldsymbol{z}_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}\right), \qquad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\mathsf{T}$$

Marginalising

- We want $p(\boldsymbol{z}_1) = \int p(\boldsymbol{z}_1, \boldsymbol{z}_2)\,\mathrm{d}\boldsymbol{z}_2$
- Doing the integral directly takes a page of block-matrix algebra (see the notes)
- Instead take $\boldsymbol{t} = (\boldsymbol{t}_1, \boldsymbol{0})$ in the MGF, so that $\boldsymbol{t}^\mathsf{T}\boldsymbol{z} = \boldsymbol{t}_1^\mathsf{T}\boldsymbol{z}_1$

$$\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}_1^\mathsf{T}\boldsymbol{z}_1}\right] = \mathrm{e}^{\boldsymbol{t}_1^\mathsf{T}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{t}_1^\mathsf{T}\boldsymbol{\Sigma}_{11}\boldsymbol{t}_1}$$

- That is the MGF of $\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$, so

$$\boldsymbol{z}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$$

- To marginalise, just strike out the rows and columns of the variables you integrate away

Conditioning: a Light Derivation

- We want $p(\boldsymbol{z}_1 \mid \boldsymbol{z}_2)$
- Subtract the best linear prediction of $\boldsymbol{z}_1$ from $\boldsymbol{z}_2$

$$\boldsymbol{w} = \boldsymbol{z}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{z}_2$$

- $\boldsymbol{w}$ is a linear map of $\boldsymbol{z}$, so $(\boldsymbol{w}, \boldsymbol{z}_2)$ are jointly Gaussian, and

$$\mathrm{Cov}(\boldsymbol{w}, \boldsymbol{z}_2) = \boldsymbol{\Sigma}_{12} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{22} = \boldsymbol{0}$$

- For jointly Gaussian variables, uncorrelated means independent (the MGF factorises)
- So knowing $\boldsymbol{z}_2$ tells us nothing about $\boldsymbol{w}$, and $\boldsymbol{z}_1 = \boldsymbol{w} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{z}_2$ with $\boldsymbol{z}_2$ now fixed

The Conditional Distribution

- From the linear-transformation rule, $\boldsymbol{w}$ has mean $\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2$ and covariance $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$
- Hence

$$\boldsymbol{z}_1 \mid \boldsymbol{z}_2 \sim \mathcal{N}\big(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{z}_2 - \boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big)$$

- The mean moves linearly with what we observe
- The covariance shrinks, by an amount that does not depend on the value we observe
- This one formula is the engine behind Gaussian processes and Kalman filters

The Long Way Round

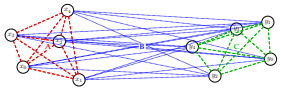
- The same results follow by brute force from the block factorisation

$$\begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ \boldsymbol{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \boldsymbol{S} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \boldsymbol{0} \\ \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{I} \end{pmatrix}$$

  where $\boldsymbol{S} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ (the **Schur complement**)
- Inverting each factor splits the exponent into a part for $\boldsymbol{z}_2$ and a part for $\boldsymbol{z}_1$ given $\boldsymbol{z}_2$
- It also gives $\det\boldsymbol{\Sigma} = \det\boldsymbol{S}\det\boldsymbol{\Sigma}_{22}$
- The full derivation is in the notes: worth reading once, not memorising

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A Noisy Measurement

- Prior belief about a quantity: $x \sim \mathcal{N}(\mu_0, s_0^2)$
- We measure $y = x + \epsilon$ with noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- (x,y) is a linear map of (x,ϵ), so jointly Gaussian with

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_0 \\ \mu_0 \end{pmatrix}, \begin{pmatrix} s_0^2 & s_0^2 \\ s_0^2 & s_0^2 + \sigma^2 \end{pmatrix} \right)$$

- Conditioning on y

$$\mathbb{E}[x \mid y] = \mu_0 + \frac{s_0^2}{s_0^2 + \sigma^2}(y - \mu_0), \qquad \frac{1}{\mathbb{V}\mathrm{ar}(x \mid y)} = \frac{1}{s_0^2} + \frac{1}{\sigma^2}$$

- The estimate moves towards the measurement, and the precisions (inverse variances) add

Combining Sensors

- Two sensors measure the same x, with noise variances σ_1^2 and σ_2^2, giving y_1 and y_2
- With a flat prior ($s_0 \to \infty$), applying the last result twice gives

$$\mathbb{E}[x \mid y_1, y_2] = \frac{y_1/\sigma_1^2 + y_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}, \qquad \frac{1}{\mathbb{V}\mathrm{ar}(x \mid y_1, y_2)} = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}$$

- A precision-weighted average: trust the better sensor more
- With n equally good measurements the variance falls as σ^2/n

Predicting One Variable from Another

- Two correlated quantities (e.g. height and weight) with correlation ρ

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right)$$

- Conditioning on x_1

$$\mathbb{E}[x_2 \mid x_1] = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1), \qquad \mathbb{V}\mathrm{ar}(x_2 \mid x_1) = \sigma_2^2(1 - \rho^2)$$

- The prediction is a straight line: linear regression
- As $|\rho| < 1$ the prediction is fewer standard deviations from the mean than x_1 is: **regression to the mean**

Tracking: the Kalman Filter

- A position drifts, $x_{t+1} = x_t + \eta_t$ with $\eta_t \sim \mathcal{N}(0, q)$, and is observed as $y_t = x_t + \epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$
- Our belief stays Gaussian, $x_t \sim \mathcal{N}(m_t, v_t)$. **Predict** (a sum of Gaussians): $x_{t+1} \sim \mathcal{N}(m_t,\, v_t + q)$
- **Update:** a noisy measurement, with prior $\mathcal{N}(m_t, v_t + q)$

$$m_{t+1} = m_t + k(y_{t+1} - m_t), \qquad v_{t+1} = (1-k)(v_t + q),$$
$$k = \frac{v_t + q}{v_t + q + \sigma^2}$$

- That is the Kalman filter (with vectors and matrices it guides aircraft and phones)

Towards Gaussian Processes

- Suppose the function values at some inputs are jointly Gaussian, and we observe some of them with noise
- Put the unknown value f_* in $\boldsymbol{z}_1$ and the observations $\boldsymbol{y}$ in $\boldsymbol{z}_2$

$$\begin{pmatrix} f_* \\ \boldsymbol{y} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} k_* & \boldsymbol{k}_*^\mathsf{T} \\ \boldsymbol{k}_* & \mathbf{K} + \sigma^2 \mathbf{I} \end{pmatrix}\right)$$

- The conditional formula gives the prediction immediately

$$f_* \mid \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{y},\; k_* - \boldsymbol{k}_*^\mathsf{T}(\mathbf{K} + \sigma^2\mathbf{I})^{-1}\boldsymbol{k}_*\big)$$

- Where the covariances come from is the subject of the next lecture

Summary

- The toolkit
  - ⋆ $\int \mathrm{e}^{-\frac{1}{2}\boldsymbol{x}^\mathsf{T}\mathbf{A}\boldsymbol{x} + \boldsymbol{b}^\mathsf{T}\boldsymbol{x}} \mathrm{d}\boldsymbol{x} = \sqrt{(2\pi)^n / \det \mathbf{A}}\; \mathrm{e}^{\frac{1}{2}\boldsymbol{b}^\mathsf{T}\mathbf{A}^{-1}\boldsymbol{b}}$
  - ⋆ MGF: $\mathbb{E}\left[\mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{x}}\right] = \mathrm{e}^{\boldsymbol{t}^\mathsf{T}\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}^\mathsf{T}\boldsymbol{\Sigma}\boldsymbol{t}}$
  - ⋆ Linear maps: $\boldsymbol{W}\boldsymbol{x} + \boldsymbol{c} \sim \mathcal{N}(\boldsymbol{W}\boldsymbol{\mu} + \boldsymbol{c}, \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\mathsf{T})$
  - ⋆ Marginals: strike out rows and columns
  - ⋆ Conditionals: $\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{z}_2 - \boldsymbol{\mu}_2)$ and $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$
- Gaussians stay Gaussian, so inference reduces to linear algebra on means and covariances